

PaperReading

linzy

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Contents

1	Glueball (different method and else)	4
1.1	Gluons in glueballs: Spin or helicity	4
1.2	The glueball spectrum from an anisotropic lattice study	5
1.3	吕晓睿-郑州暑期学校	5
1.4	Calculations of Glueball Wavefunction in Lattice QCD	5
1.5	Glueball phenomenology and the relativistic flux tube model	5
1.6	Glueball Calculations in Large-Nc Gauge Theory	5
1.7	Pure glueball states in a Light-Front holographic approach	6
1.8	On Glueballs, Pomeron, and Odderon in Holographic Quantum Chromodynamics: From Hadron Spectroscopy to Collider Phenomenology	6
1.9	Discovery of a Glueball-like particle X(2370) at BESIII	6
1.10	The Glueball in a Chiral Linear Sigma Model with Vector Mesons	6
1.11	Dispersive analysis of glueball masses	6
1.12	Light Scalar Isoscalar Mesons and the Glueball	7
1.13	SCALAR GLUEBALL: ANALYSIS OF THE (IJP C = 00++) WAVE	7
1.14	Revisiting Scalar Glueballs	7
1.15	Scalar Mesons below 1 GeV	7
1.16	Scalar mesons above and below 1 GeV	8
1.17	Scalar isoscalar mesons and the scalar glueball from radiative J/ ψ decays	8
1.18	Glueballs from bound state equations	8
1.19	Glueballs in Radiative J/ ψ Decays	8
1.20	Glueballs from the Bethe-Salpeter equation	8
1.21	Two- and Three-gluon Glueballs within the Helicity Formalism	8
1.22	Semirelativistic potential model for three-gluon glueballs	8
1.23	Lattice Evidence that Scalar Glueballs Are Small	9
1.24	Glueballs, Constituent Gluons and Instantons	9
2	Glueball (mass spectrum)	9
2.1	Exotic glueball $0^{\pm-}$ states in QCD Sum Rules	9
2.2	Spectra of glueballs and oddballs and the equation of state from holographic QCD	9
2.3	The low lying glueball spectrum	9

3	Glueball (review)	10
3.1	A review on Glueball hunting	10
3.2	The Status of Glueballs.	10
3.3	Mass creation by the strong interaction: Glueballs - status and perspectives	10
3.4	Glueballs, a fulfilled promise of QCD?	10
3.5	Glueballs as the Ithaca of meson spectroscopy	10
4	Glueball (color and Cparity)	10
4.1	Constituent gluon interpretation of glueballs and gluelumps	10
4.2	Qualitative features of the glueball spectrum	11
4.3	Psi Resonances, Gluons and the Zweig Rule	11
5	Glueball(experiment)	11
5.1	X(2370) glueball-like particle productions at the BESIII	11
5.2	Highlights of light meson spectroscopy at the BESIII experiment	11
6	Glueball (Candidate)	11
6.1	Is $f_0(1500)$ a scalar glueball?	11
6.2	Evidence for a scalar glueball	11
6.3	Revisiting Scalar Glueballs	12
6.4	Measurements of J/ψ decays into $\omega K \bar{K} \pi$ and $\phi K \bar{K} \pi$	12
6.5	Observation of pseudoscalar and tensor resonances in $J/\psi \rightarrow \gamma \phi \phi$	12
6.6	Determination of Spin-Parity Quantum Numbers of $X(2370)$ as 0^{-+}	12
7	Decay	12
7.1	Exclusive diffractive processes at HERA within the dipole picture	12
7.2	Masses, decays and mixings of gluonia in QCD	13
8	QCD Theory	13
8.1	Symmetries of baryons and mesons	13
8.2	Color in QCD	13
9	Structure Function	13
9.1	Polarized Parton Distributions and Future Neutrino Factories	13
9.2	Parton Distribution Functions and their Generalizations	13
9.3	Nucleon structure from basis light-front quantization	14
10	GFF	14
10.1	Gravitational form factors and mechanical properties of quarks in protons: A basis light-front quantization approach	14
10.2	Gravitational form factors of glueballs in Yang-Mills theory	14
10.3	Gravitational form factors of charmonia	15
10.4	Lattice evidence that scalar glueballs are small	15

10.5	Light-Cone Representation of the Spin and Orbital Angular Momentum of Relativistic Composite Systems	15
10.6	Gravitational form factor D of charmonium from shear stress	15
11	Review	15
11.1	Light front QCD in a transverse harmonic oscillator basis	15
11.2	A Friendly Introduction to the Light Front	15
11.3	Nucleon Structure from Basis Light-Front Quantization : Status and Prospects . .	16
11.4	Field theories in the infinite-momentum frame	16
12	DEMELO Suggestion	16
12.1	Covariance of light-front models: pair current	16
12.2	Pairs in the light-front and covariance	16
12.3	Light-front projection of spin-1 electromagnetic current and zero-modes	16
12.4	Covariant and light-front approaches to the ρ -meson electromagnetic form factors .	17
12.5	Pion electromagnetic current in a light-front model	17
12.6	Unambiguous extraction of the electromagnetic form factors for spin-1 particles on the light-front	17
12.7	Ward-Takahashi Identity on the Light Front	17
13	blfq method	17
13.1	Transverse momentum structure of strange and charmed baryons: a light-front Hamiltonian approach	17
13.2	Transverse structure of the pion beyond leading twist with basis light-front quan- tization	17
13.3	Heavy Quarkonium in a Holographic Basis	18
14	AI4Science	18
14.1	Heavy Quarkonium Spectrum and Decay Constants from a Neural-Network-Based Holographic Model	18

1 Glueball (different method and else)

1.1 Gluons in glueballs: Spin or helicity

Title: Gluons in glueballs: Spin or helicity. [42]

这篇文章提出了一些新的关于胶子自旋的处理方法的讨论，通过它的讨论结果构造了一种哈密顿量模型，并得到了两胶子体系的能谱结构，并将其与格点的结果进行比较。值得一提的是，这篇文章中提到了关于瞬时相互作用对于哈密顿量的不可或缺的意义。

这篇文章中提到了一些关于胶球系统和QCD等的一些基础概念：

Like other effective approaches of QCD, potential models still have difficulties to cope with gluonic hadrons. Assuming that glueballs are bound states of valence gluons with zero current mass, it is readily understood that the use of a potential model, intrinsically non covariant, could be problematic in this case.

在处理涉及到胶子的强子时，很多有效模型都遇到了困难。这里还提到了一点有趣的结论：胶球是0流质量的价胶子。

The main challenge for this kind of model is actually to find a way to introduce properly the more relevant degree of freedom of the gluon: spin or helicity.

对于胶子来说，描述它的关于自旋的自由度选择自旋还是螺旋度是一个很关键的问题。

As quantum chromodynamics (QCD) is built on the nonabelian SU(3)-color group, it allows for purely gluonic bound states called glueballs.

量子色动力学建立在SU(3)的颜色群上，所以允许一种仅包含胶子的基态，叫做胶球 (glueball)。

On the other hand, pure gauge QCD has been investigated by lattice QCD for many years, leading to a well established glueball spectrum below 4 GeV [2, 3].

在这篇文章的参考文献2,3中，介绍了由格点计算的胶球能谱。

As a bound state of two gluons can only have $C = +$, it is rather natural to assume that the lightest glueballs are mainly two-gluon states (the more constituent gluons are present, the more the glueball should be heavy).

这里提到了很重要的一点，关于胶子系统的宇称。由于宇称的限制，最轻的胶子系统应当包含两个胶子。但是关于如何确定整个系统的 J^{PC} 宇称，这是一个非常重要但是复杂的问题，甚至我认为这作为一个量子数应当在构建基态时就进行处理，需要进一步对其进行研究。

Moreover, it is interesting to mention some results of the Coulomb gauge study of Ref. [8]. In this approach, a Fock space expansion of glueball states in terms of quasigluons can be performed, and it appears that the influence of the three- and four-gluon components on the low-lying $C = +$ glueballs is negligible: The two-gluon component is dominant as intuitively expected.

对于在福克空间中展开的胶球系统，他们采取库伦规范。为什么有这样的要求？在参考文献8中也许有这个问题的答案。在福克空间展开中，三胶子，四胶子组分仅占了很小的比例，主要是二胶子的系统。

In this picture, the valence gluon is a posteriori massive, because it is confined into a glueball. Let us note that, more generally, both quarks and gluons can gain a constituent mass from renormalization theory (in the Coulomb gauge approach of Ref. [14], massless gluons gain a constituent mass of about 0.7 GeV at zero momentum). But, other studies keep the assumptions

of Ref. [10] and state that a valence gluon has to be a priori considered as massive, that is with a non zero current mass [10, 15].

在参考文献10,14,15中提出了对于胶子质量的两种不同的处理方式，也是本文的核心讨论部分。

Moreover, the glueball spectrum is now far better known than at the time of Ref. [9] thanks to lattice QCD calculations [2, 3].

参考文献2,3,9中介绍了格点计算胶球质量谱的结果。

It is of great phenomenological interest to be able to express a given helicity state in terms of states of given orbital angular momentum L and intrinsic spin S .

关于角动量和自旋的波函数是一个非常引人关注的点。这篇文章中同样花了较小的篇幅介绍相关波函数的处理和作用，但是并没有给出计算结果。

In particular, the necessity of adding instanton-induced forces should not be seen as a flaw of the model, but rather as a way to be more coherent with other studies showing that instantons contribute in glueballs.

加入瞬时相互作用项是一种更接近物理的要求而非单纯的模型上的调整。

1.2 The glueball spectrum from an anisotropic lattice study

Title: The glueball spectrum from an anisotropic lattice study. [44]

1.3 吕晓睿-郑州暑期学校

PPT中介绍了一些BESIII的实验，其中包括了 J/Ψ 的衰变中可能的与胶球相关的部分。(P31)

https://indico.ihep.ac.cn/event/24901/contributions/199839/attachments/94226/123556/0824%E5%90%95%E6%99%93%E7%9D%BF-LYU_%E9%83%91%E5%B7%9E%E8%AE%B2%E4%B9%A0%E7%8F%AD2025.pdf

BESIII的介绍，其中有关于胶球的实验的详细的的信息。

https://indico.ihep.ac.cn/event/19694/contributions/133803/attachments/69026/82586/BJLiu_BESIII500.pdf

1.4 Calculations of Glueball Wavefunction in Lattice QCD

Title: Calculations of Glueball Wavefunction in Lattice QCD [54].

格点计算胶球的波函数的文章。

1.5 Glueball phenomenology and the relativistic flux tube model

Title: Glueball phenomenology and the relativistic flux tube model[12].

1.6 Glueball Calculations in Large-Nc Gauge Theory

Title: Glueball Calculations in Large-Nc Gauge Theory [18].

用光前方法计算胶球的的文章。

1.7 Pure glueball states in a Light-Front holographic approach

Title: Pure glueball states in a Light-Front holographic approach [50]

Satvir推荐的glueball的文章，它们采用了Light-Front holographic的方法进行计算。

其中提到了Lattice的结果， 0^{++} 也是在1710附近的。然而soft wall model等会将 $f_0(500)$ 作为基态进行fit。

还有一篇专门计算标量胶球的内容类似的文章：<https://arxiv.org/pdf/2002.11720>

1.8 On Glueballs, Pomeron, and Odderon in Holographic Quantum Chromodynamics: From Hadron Spectroscopy to Collider Phenomenology

Title: On Glueballs, Pomeron, and Odderon in Holographic Quantum Chromodynamics: From Hadron Spectroscopy to Collider Phenomenology[29]

Satvir推荐的关于glueball的综述文章，其中详细介绍了如何通过HolographicQCD计算胶球的过程。

基于QCD的Holographic方法，被称为ADS/QCD模型。它希望在一个五维时空里做一个超对称的理论，并且将其在QCD的思维时空中尽可能的组装起来。根据破坏共形性质方法的不同，可以区分其中的两种主要模型：hard-wall模型和 soft-wall 模型。这种方法得到的结果也适合用来计算FF, PDF,GPD等结构函数。实验上对相关结果的支持主要来自于小x区域的深度非弹实验。

1.9 Discovery of a Glueball-like particle X(2370) at BESIII

Title: Discovery of a Glueball-like particle X(2370) at BESIII [30]

天才推荐的文章。tensor胶球在BESIII的实验中得到了一些可能的数据，大多数是来自于J粒子的衰变。

还可以看到关于这篇文章的工作的一些汇报：

<https://indico.ihep.ac.cn/event/23322/contributions/166226/attachments/86033/110266/7th%20Prof.%20Jin%20X2370-J50.pdf>

1.10 The Glueball in a Chiral Linear Sigma Model with Vector Mesons

Title: The Glueball in a Chiral Linear Sigma Model with Vector Mesons [33]

线性 σ 模型对胶球的计算。这里它们在研究 $f_0(1370), f_0(1500), f_0(1710)$ 等。这篇文章的重点在于 σ 模型，这可能和自发对称性破缺有关系。后面还需要仔细阅读模型相关的介绍。

1.11 Dispersive analysis of glueball masses

Title:Dispersive analysis of glueball masses [38]

利用算符乘积展开（OPE）计算glueball能谱结构的文章。在0.6GeV处发现了一个较小的峰，它认为这是最轻的meson： $f_0(500)$ 中含有较少的胶球的成分。

1.12 Light Scalar Isoscalar Mesons and the Glueball

Title: Light Scalar Isoscalar Mesons and the Glueball [25]

对f系列的介子进行详细分析的文章。

Convention部分介绍了f的定义: Thus the names for scalar isoscalar mesons which consist of a quark and an antiquark ($q\bar{q}$) and which have the quantum numbers of the vacuum, i.e. spin $J = 0$, parity $P = +$, charge conjugation-parity $C = +1$ and isospin $I = 0$ are as follows:

If the object is composed of $u\bar{u} + d\bar{d}$, or $s\bar{s}$ or mixtures of the two components it should be called f_0, f'_0

$f_0(500)$ 的宽度非常大, 难以获得一个关于它的准确的质量。这一点可以在这篇文章的fig1中很直观的体现出来。

In the quark model by Gell-Mann [2] and Zweig [3] two isoscalar quark-antiquark ($q\bar{q}$) states are expected: $u\bar{u} + d\bar{d}$ and $s\bar{s}$. The physical states are mixtures of these two states with an unknown mixing angle.

这个观点可能比较有启发性, 对于轻介子的物理态, 它们是两种理论上的态以某种混合角混合得到的。那么对于胶球而言, 我们需要证明它异于上面说的混合态。

The predictions have converged during the past decade to a scalar glueball ($JPC = 0^{++}$) with a mass of 1.6 GeV/c² as lowest state.

从05年开始似乎从Lattice到Sum rule的方法就不认为有很低质量的胶球了, 普遍认为都在1.5GeV以上。那么 $f_0(500)$ 似乎不能算是胶球?

Having tentatively merged $f_0(600)$ and $f_0(1370)$ into one resonance, one realizes that a monster resonance has been created with a mass and width of about 800 MeV/c²

这篇文章最后认为它可能 $f_0(500)$ 和它的态可以合并为一个新的共振态。

1.13 SCALAR GLUEBALL: ANALYSIS OF THE ($JPC = 0^{++}$) WAVE

Title: SCALAR GLUEBALL: ANALYSIS OF THE ($JPC = 0^{++}$) WAVE [9]

1997年的文章, 使用了propagator matrix (D-matrix) technique, 认为 $f_0(1300)$, $f_0(1500)$ 是一个可能的候选态。

1.14 Revisiting Scalar Glueballs

Title: Revisiting Scalar Glueballs [14]

15年的文章, 支持1710是可能的基态。

整理了到15年为止Lattice对胶球的计算

1.15 Scalar Mesons below 1 GeV

Title: Scalar Mesons below 1 GeV

<https://pdg.lbl.gov/2023/reviews/rpp2023-rev-scalar-mesons.pdf#:~:text=In%20this%20note%2C%20we%20discuss%20the%20light%20scalars,and%20f0%28980%29%20Abstract%20both%20with%20I%20%3D%200.>

PDG中详细介绍标量介子, 包括 $f_0(500)$ 的文章。

1.16 Scalar mesons above and below 1 GeV

Title: Scalar mesons above and below 1 GeV [17]

1.17 Scalar isoscalar mesons and the scalar glueball from radiative J/ψ decays

Title: Scalar isoscalar mesons and the scalar glueball from radiative J/ψ decays [52]

主要是介绍BESIII中的J/ψ衰变的实验数据的文章。

Introduction部分写得非常好。有关于f介子的介绍和胶球历代各种方法的理论计算。

1.18 Glueballs from bound state equations

Title: Glueballs from bound state equations [31]

BSE方法计算胶球基态的文章，2022年的比较新并且和格点的结果有比较。

1.19 Glueballs in Radiative J/ψ Decays

Title: Glueballs in Radiative J/ψ Decays [36]

分析BESIII的衰变的实验数据的，值得细看。

这里提到了胶球和拥有相同的量子数的介子会发生混合，可以通过下面的这种混合矩阵描述，通过衰变的数据可以分析得到具体的矩阵元。

$$\begin{pmatrix} f_0(1370) \\ f_0(1500) \\ f_0(1710) \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} \begin{pmatrix} |n\bar{n}\rangle \\ |s\bar{s}\rangle \\ |gg\rangle \end{pmatrix}$$

1.20 Glueballs from the Bethe-Salpeter equation

Title: Glueballs from the Bethe-Salpeter equation[51]

Bethe-Salpeter equation (BSE)计算胶球。得到的结果在1.6左右。可以用来了解这个方法。和DSE可能有一些联系。

1.21 Two- and Three-gluon Glueballs within the Helicity Formalism

Title: Two and Threegluon Glueballs within the Helicity Formalism [16]

用Helicity Formalism计算胶球能谱的文章。其中有较深入的对于C宇称的讨论。

1.22 Semirelativistic potential model for three-gluon glueballs

Title: Semirelativistic potential model for threegluon glueballs[43]

计算胶球的哈密顿量方法。给的信息比较少，但是引用中可能有一些和wavefunction相关的结果。

1.23 Lattice Evidence that Scalar Glueballs Are Small

Title: Lattice Evidence that Scalar Glueballs Are Small [1]

近期发表在PRL上的文章，用格点计算了胶球的GFF和半径。

1.24 Glueballs, Constituent Gluons and Instantons

Title: Glueballs, Constituent Gluons and Instantons [53]

Satvir推荐的glueball的文章，计算的态挺全的，另外也尝试了一些花活。

重点：这是一篇基于 BSE（Bethe-Salpeter / Salpeter 方程）的工作。文章在纯 Yang-Mills 理论下构建了组分二胶球的图像，并通过对角化哈密顿量/求解约化的 Bethe-Salpeter (Salpeter) 方程得到胶球谱。其内核包含 Casimir 标度的伴随禁闭、短程伴随库伦相互作用，以及由 instanton 诱导的中心力和张量力。

重点：Instanton 对不同 J^{PC} 通道的影响有显著差异。具体而言：

- 0^{++} （标量）道：instanton 诱导出很强的吸引力，使得标量胶球异常紧凑，其半径量级为 instanton 大小 $\rho \sim \frac{1}{3}$ fm，与格点的指示一致；
- 0^{-+} （赝标量）道：相应的 instanton 力是排斥的；
- 2^{++} （张量）道：instanton 力几乎不存在，态由于离心势垒保持空间延展。

这一图像与早期 Schäfer-Shuryak 的经典结论一致，也是理解标量胶球“反常紧凑”的关键物理机制。文章同时讨论了 S - D 混合，并给出激发态的半经典 Regge 行为分析。

2 Glueball (mass spectrum)

2.1 Exotic glueball $0^{\pm-}$ states in QCD Sum Rules

Title: Exotic glueball $0^{\pm-}$ states in QCD Sum Rules [49]

基于QCD Sumrule 的计算C字称为负的态的文章。但是结果有点太夸张了。

2.2 Spectra of glueballs and oddballs and the equation of state from holographic QCD

Title: Spectra of glueballs and oddballs and the equation of state from holographic QCD [60]

21年的文章，holographic方法计算C字称为-的胶球的质量。Table2中给出了全面的能谱的整理。

2.3 The low lying glueball spectrum

Title: THE LOW LYING GLUEBALL SPECTUM [55]

同样是QCD Hamiltonian的方法，另外Intro写得不错。

3 Glueball (review)

3.1 A review on Glueball hunting

Title: A review on Glueball hunting[56]

23年的综述文章。从算符层面进行了分析，也介绍了BSE,Lattice的一些结果，最后还做了一些实验数据的总结。

3.2 The Status of Glueballs.

Title: The Status of Glueballs. [48]

3.3 Mass creation by the strong interaction: Glueballs - status and perspectives

Title: Mass creation by the strong interaction: Glueballs - status and perspectives [57]

这篇文章介绍了目前的能谱，也讨论了如何在实验上寻找胶球。

3.4 Glueballs, a fulfilled promise of QCD?

Title: Glueballs, a fulfilled promise of QCD? [35]

这篇文章是 "50 Years of Quantum Chromodynamics" 综述中的一章。回顾了胶球寻找的历史，阐述了如何基于胶球的特性（特别是 J/ψ 辐射衰变中的超额产生）来寻找它。其核心结果来自一个耦合道分波分析，给出了标量胶球的证据和张量胶球的初步线索。此外， $\psi(2S)$ 和 $\Upsilon(1S)$ 的辐射衰变数据也显示出可能来自胶球产生的标量强度。

3.5 Glueballs as the Ithaca of meson spectroscopy

Title: Glueballs as the Ithaca of meson spectroscopy [40]

一篇关于 gluonium（胶球）的紧凑综述，涵盖了 J/ψ 辐射衰变、矢量介子衰变、标量介子混合、通过 Pomeron 和 Odderon 的高能截面、胶球衰变等多个方面。积累的证据支持 $f_0(1710)$ 在标量介子中具有最大（也是最大的）胶球组分，但没有任何单一观测量是决定性的。赝标量和张量胶球还没有明确的实验候选者。

4 Glueball (color and Cparity)

4.1 Constituent gluon interpretation of glueballs and gluelumps

Title: Constituent gluon interpretation of glueballs and gluelumps [10]

对2, 3, 4个胶子组成的胶球的颜色空间进行了仔细的分析，并给出了每个色单态给出的C宇称。

4.2 Qualitative features of the glueball spectrum

Title: Qualitative features of the glueball spectrum [32]

很老的文章，从算符层面分析了胶球的可能的量子数。

4.3 Psi Resonances, Gluons and the Zweig Rule

Title: Psi Resonances, Gluons and the Zweig Rule [27]

非常老的文章，被引很高。分析了胶球各种不同的量子数。

5 Glueball(experiment)

5.1 X(2370) glueball-like particle productions at the BESIII

Title: X(2370) glueball-like particle productions at the BESIII [13]

这篇是2370的glueball like, 还有一篇差不多的发表在PRL上，但是那篇没有讲明glueball的事。

5.2 Highlights of light meson spectroscopy at the BESIII experiment

Title: Highlights of light meson spectroscopy at the BESIII experiment [34]

BESIII实验的文章，记录的比较全面，其中也介绍了胶球相关的信息。

6 Glueball (Candidate)

6.1 Is $f_0(1500)$ a scalar glueball?

Title: Is $f_0(1500)$ a scalar glueball? [8]

在CERN的LEAR上发现了两个新的标量介子 $f_0(1370)$ 和 $f_0(1500)$ 之后，作者论证了这两个态不可能都是 $Q\bar{Q}$ 介子。文章指出 $f_0(1500)$ 的性质与一个位于1500 MeV附近的基态胶球和附近的 0^{++} $Q\bar{Q}$ 九重态混合后的结果一致。该假设的一个可检验的预言是存在一个附加的标量态 $f'_0(1500-1800)$ ，它与 $K\bar{K}$ 、 $\eta\eta$ 和 $\eta\eta'$ 强耦合。文章还讨论了在 ~ 2 GeV处张量胶球的可能信号。这篇是胶球候选态研究的经典文献之一。

6.2 Evidence for a scalar glueball

Title: Evidence for a scalar glueball [7]

这是上一篇 [8] 的早期版本 (Phys. Lett. B)，作者认为LEAR上新发现的 $f_0(1500)$ 其性质与格点QCD预言的最轻标量胶球相符，而与 $Q\bar{Q}$ 态不符。胶球衰变可以通过与附近 $Q\bar{Q}$ 态的混合，或衰变到一对胶球（包括 η 、 η' 或 $(\pi\pi)_S$ ）来理解。 $f_0(1500)$ 的偏宽度与这一图像一致，并给出可检验的辐射衰变预言 $\gamma\phi$ 、 $\gamma\omega$ 、 $\gamma\rho$ ，以及一个附加的 f'_0 态。

6.3 Revisiting Scalar Glueballs

Title: Revisiting Scalar Glueballs [15]

普遍认为最轻的标量胶球隐身在 $f_0(1370)$ 、 $f_0(1500)$ 和 $f_0(1710)$ 等同位旋标量介子中。文章综合格点QCD（包括淬灭和非淬灭）和实验数据，推断这些 f_0 态的胶球和 $q\bar{q}$ 成分。考虑的观测量包括：标量胶球质量的格点计算、辐射衰变 $J/\psi \rightarrow \gamma f_0$ 的测量、 f_0 衰变到 $\pi\pi$ 、 $K\bar{K}$ 和 $\eta\eta$ 的分支比、 J/ψ 衰变到 $f_0(1710)\omega$ 与 $f_0(1710)\phi$ 的比值、 f_0 对 $B_s \rightarrow J/\psi \pi^+ \pi^-$ 的贡献，以及 $a_0(1450)$ 和 $K_0^*(1430)$ 的近简并。所有分析都指向 $f_0(1710)$ 具有显著的胶球性质，而 $f_0(1500)$ 具有味道八重态结构。

6.4 Measurements of J/ψ decays into $\omega K \bar{K} \pi$ and $\phi K \bar{K} \pi$

Title: Measurements of J/ψ decays into $\omega K \bar{K} \pi$ and $\phi K \bar{K} \pi$ [4]

BESII利用BEPC上收集的 5.8×10^7 个 J/ψ 事例研究了 $J/\psi \rightarrow \omega K \bar{K} \pi$ 和 $J/\psi \rightarrow \phi K \bar{K} \pi$ 衰变。在 $J/\psi \rightarrow \omega K \bar{K} \pi$ 产生的 $K_S^0 K^\pm \pi^\mp$ 和 $K^+ K^- \pi^0$ 系统中，不变质量分布在 $1.44 \text{ GeV}/c^2$ 附近看到增强，这可能与某些标量胶球候选态相关；而 $J/\psi \rightarrow \phi K \bar{K} \pi$ 的 $K \bar{K} \pi$ 系统中没有观察到质量增强。文章测量了 $J/\psi \rightarrow \omega K_S^0 K^\pm \pi^\mp$ 、 $\phi K_S^0 K^\pm \pi^\mp$ 、 $\omega K^* \bar{K} + \text{c.c.}$ 、 $\phi K^* \bar{K} + \text{c.c.}$ 的分支比，并首次测量了 $J/\psi \rightarrow \eta K_S^0 K^\pm \pi^\mp$ 的分支比。

6.5 Observation of pseudoscalar and tensor resonances in $J/\psi \rightarrow \gamma \phi \phi$

Title: Observation of pseudoscalar and tensor resonances in $J/\psi \rightarrow \gamma \phi \phi$ [5]

BESIII基于BEPCII上收集的 $(1310.6 \pm 10.5) \times 10^6$ 个 J/ψ 事例，对 $J/\psi \rightarrow \gamma \phi \phi$ 进行了分波分析。PWA结果显示结构主要是 0^{-+} 态：确认了 $\eta(2225)$ 的存在并测量了其共振参数；观察到两个新的赝标量态 $\eta(2100)$ （质量约 $2050 \text{ MeV}/c^2$ ）和 $X(2500)$ （质量约 $2470 \text{ MeV}/c^2$ ）。此外还观察到了标量态 $f_0(2100)$ 以及三个张量态 $f_2(2010)$ 、 $f_2(2300)$ 、 $f_2(2340)$ 。文章指出 $f_2(2340)$ 在 J/ψ 辐射衰变中的较大产生率与格点QCD对张量胶球的预言相符，是 0^{-+} 和 2^{++} 胶球候选态的重要实验线索。

6.6 Determination of Spin-Parity Quantum Numbers of $X(2370)$ as 0^{-+}

Title: Determination of Spin-Parity Quantum Numbers of $X(2370)$ as 0^{-+} from $J/\psi \rightarrow \gamma K_S^0 K_S^0 \eta'$ [6]

BESIII基于 $(10087 \pm 44) \times 10^6$ 个 J/ψ 事例，对 $J/\psi \rightarrow \gamma K_S^0 K_S^0 \eta'$ 进行了分波分析。测量得到 $X(2370)$ 的质量为 $2395 \pm 11(\text{stat})_{-94}^{+26}(\text{syst}) \text{ MeV}/c^2$ ，宽度为 $188_{-17}^{+18}(\text{stat})_{-33}^{+124}(\text{syst}) \text{ MeV}$ ，统计显著性大于 11.7σ 。最重要的结果是首次将 $X(2370)$ 的自旋宇称确定为 0^{-+} 。其质量和 J^{PC} 都与格点QCD预言的最轻赝标量胶球（预期质量约 $2.395 \pm 0.014 \text{ GeV}/c^2$ ）相符，使 $X(2370)$ 成为目前最受关注的赝标量胶球候选态之一。

7 Decay

7.1 Exclusive diffractive processes at HERA within the dipole picture

Title: Exclusive diffractive processes at HERA within the dipole picture [37]

这篇是从波函数计算衰变的文章，小毅的工作借鉴了这篇。但是衰变的过程和我想研究的很不一样。

7.2 Masses, decays and mixings of gluonia in QCD

Title: Masses, decays and mixings of gluonia in QCD [46]

利用QCD谱求和规则 (QCD spectral sum rules) 和低能定理计算了胶球 (gluonia) 的质量和衰变宽度。在标量道, 得到一个质量 $M_G = (1.5 \pm 0.2)$ GeV 的胶球, 主要衰变到 $U(1)_A$ 道 $\eta\eta'$ 和 $4\pi^0$ 。同时需要引入与夸克自由度强耦合的宽低质量胶球 (σ_B 及其径向激发) 来保证整个方法的一致性。结合 $\bar{q}q$ 夸克态的结果, 提出了最大胶球-夸克混合方案, 很好地描述了标量介子谱 $\sigma(1.0)$ 、 $f_0(0.98)$ 、 $f_0(1.37)$ 、 $f_0(1.5)$ 和 $f_J(1.71)$ 。在张量道, 胶球质量 $M_T \simeq (2.0 \pm 0.1)$ GeV, 认为 $\zeta(2.2)$ 是 2^{++} 胶球候选者。在赝标量道, 胶球质量 $M_P \simeq (2.05 \pm 0.19)$ GeV, 并指出 $E/\iota(1.44)$ 与胶子流的耦合弱于 η' , 支持其作为 $\eta'(0.96)$ 的第一径激发的解释。

8 QCD Theory

8.1 Symmetries of baryons and mesons

Title: Symmetries of baryons and mesons [28]

在 R. D. Field - Applications of Perturbative QCD 的附录D中引用了这篇文章, 其中包含了最早的关于 colorfactor 的讨论。

文章中并未具体介绍如何计算某一个 ColorFactor, 这里讨论了重子和介子中存在的群结构。其内容接近于国科大粒子物理基础课程中对于对称性的介绍, 认为系统中存在 $U(3) \times U(3)$ 的对称性。

不是特别有用的文章, 等有空以后再研究其技术细节。

8.2 Color in QCD

两个介绍具体颜色计算过程的PPT:

<https://www.hep.phy.cam.ac.uk/~thomson/lectures/QCD.pdf>

https://www.roma1.infn.it/people/luci/I2P/Colour_QCD.pdf

9 Structure Function

9.1 Polarized Parton Distributions and Future Neutrino Factories

Title: Polarized Parton Distributions from Charged-Current Deep-Inelastic Scattering and Future Neutrino Factories [26]

里面是关于PDF的讨论, 重要的是详细的介绍了强子张量。

9.2 Parton Distribution Functions and their Generalizations

Title: Parton Distribution Functions and their Generalizations [41].

李阳推荐的综述, 介绍了, PDF, TME, GPD 的一些信息。

9.3 Nucleon structure from basis light-front quantization

Title: Nucleon structure from basis light-front quantization[59]

记录了BLFQ框架下计算各种结构函数的方式。

10 GFF

10.1 Gravitational form factors and mechanical properties of quarks in protons: A basis light-front quantization approach

Title: Gravitational form factors and mechanical properties of quarks in protons: A basis light-front quantization approach [45]

Raj介绍的如何从波函数计算得到GFF的文章。

QCD中规范不变的张量？EMT的形式如下：

$$\begin{aligned}\theta^{\mu\nu} = & \frac{1}{2} \bar{\psi} i [\gamma^\mu D^\nu + \gamma^\nu D^\mu] \psi - F^{\mu\lambda a} F_{\lambda a}^\nu + \frac{1}{4} g^{\mu\nu} (F_{\lambda\sigma a})^2 \\ & - g^{\mu\nu} \bar{\psi} (i\gamma^\lambda D_\lambda - m) \psi,\end{aligned}\tag{1}$$

将这个张量放入态矢中可以得到下面的形式：

$$\begin{aligned}\langle P', \Lambda' | \theta_i^{\mu\nu}(0) | P, \Lambda \rangle = & \bar{u}(P', \Lambda') \left[-B_i(Q^2) \frac{\bar{P}^\mu \bar{P}^\nu}{M} + (A_i(Q^2) \right. \\ & + B_i(Q^2)) \frac{1}{2} (\gamma^\mu \bar{P}^\nu + \gamma^\nu \bar{P}^\mu) \\ & + C_i(Q^2) \frac{q^\mu q^\nu - q^2 g^{\mu\nu}}{M} \\ & \left. + \bar{C}_i(Q^2) M g^{\mu\nu} \right] u(P, \Lambda)\end{aligned}\tag{2}$$

其中A的部分就是GFF。态矢可以在blfq的基矢上展开，最终可以得到由blfq方法计算得到的GFF。

10.2 Gravitational form factors of glueballs in Yang-Mills theory

Title: Gravitational form factors of glueballs in Yang-Mills theory [2]

Nowadays, there are multiple experimental candidates for various glueball states with allowed quantum numbers.

目前胶球实验上的候选态有很多。

Information about the internal structure of hadrons may allow classification of observed hadron states as glueball-like objects. Features like their radius, or the momentum fraction carried by gluons, could serve as smoking-gun evidence of a hadron having predominantly gluonic degrees of freedom. Both of these quantities, as well as additional information like the energy distribution, are contained in their gravitational form factors (GFFs)

内部结构可能是确定胶球的决定性证据。GFF中包括很多相关的信息。

With a pure gauge action, the EMT contains only a gluonic contribution

$$T^{\mu\nu} = 2 \text{Tr} \left[-F_\alpha^\mu F^{\alpha\nu} + \frac{1}{4} g^{\mu\nu} F^{\alpha\beta} F_{\alpha\beta} \right]$$

对于胶球来说，EMT的形式会更简单一些。这也导致了它只有A和D两种GFF。

10.3 Gravitational form factors of charmonia

Title: Gravitational form factors of charmonia [58]

计算Charmonian的GFF文章。要作为计算胶球的文章的baseline。

GFF中的A可以由本征方程通过这种方式计算得到：

$$A(q_{\perp}^2) = \sum_n \sum_{\{s_i\}} \int [dx_i d^2 k_{i\perp}]_n \times \sum_j x_j \psi_n \left(\left\{ x_i, \vec{k}_{i\perp}, s_i \right\} \right) \psi_n \left(\left\{ x_i, \vec{k}_{i,j\perp}, s_i \right\} \right), \quad (3)$$

在原文的公式53，引用了[168]参考文献。

10.4 Lattice evidence that scalar glueballs are small

Title: Lattice evidence that scalar glueballs are small [3]

GFF可以和半径联系起来：

$$r_{\text{mass,BF}}^2 = \frac{1}{A(0)} \left[6 \frac{dA(t)}{dt} \Big|_{t=0} - \frac{3}{4m^2} (A(0) + 2D(0)) \right], \quad (4)$$

10.5 Light-Cone Representation of the Spin and Orbital Angular Momentum of Relativistic Composite Systems

Title: Light-Cone Representation of the Spin and Orbital Angular Momentum of Relativistic Composite Systems [11]

2000年最早的推导GFF的文章。

10.6 Gravitational form factor D of charmonium from shear stress

Title: Gravitational form factor D of charmonium from shear stress ??

天洋的文章，用 t^{12} 计算GFF-D。

11 Review

11.1 Light front QCD in a transverse harmonic oscillator basis

李阳的综述，介绍了很多BLFQ的细节,特别是谐振子相关的。其中第六页开始介绍了动能项的部分。

https://github.com/Linzy2002/Document/blob/main/notes_LFQCD_HObasis_v8.pdf

11.2 A Friendly Introduction to the Light Front

系列讲座，内容非常的具体和基础。以后写毕业论文用得到。 <https://indico.knu.ac.kr/event/718/>

更新了地址：

<https://indico.knu.ac.kr/category/10/>

11.3 Nucleon Structure from Basis Light-Front Quantization : Status and Prospects

Chandon写的综述文章，详细介绍了blfq的工作成果。

其中在3.1部分有关于质心系得处理方法的详细介绍：

By choosing λ_L sufficiently large and positive, we elevate the excited CM states to higher energies while preserving the low-lying physical states. The complete effective Hamiltonian for diagonalization becomes:

$$H'_{\text{eff}} = H_{\text{eff}} - \left(\sum_i \mathbf{p}_{i\perp} \right)^2 + \lambda_L (H_{\text{CM}} - 2b^2 I). \quad (5)$$

在第二章有关于福克空间展开的介绍：

The Fock space expansion of the eigenstate, given in Eq. (3), provides a natural basis for solving the LF Hamiltonian eigenvalue equation, Eq. (1). This expansion leads to an infinite hierarchy of coupled integral equations, which must be truncated for practical computation. This structure is reminiscent of the Dyson–Schwinger and Bethe–Salpeter equations in covariant formulations. The Light-Front Tamm–Dancoff Approximation (LFTDA) achieves this truncation by limiting the Fock space to a finite number of partons [267].

11.4 Field theories in the infinite-momentum frame

1971年的综述文章，小毅在看。有关于无穷大动量坐标系最基础的一些性质的介绍。但是书太老了。

12 DEMELO Suggestion

12.1 Covariance of light-front models: pair current

Title: Covariance of light-front models: pair current [23]

计算光前形式下 j^+ 分量的电磁流的文章。zero mode 给出了一定的贡献。

12.2 Pairs in the light-front and covariance

Title: Pairs in the light-front and covariance [24]

也是具体计算一些电磁流和积分的文章。

12.3 Light-front projection of spin-1 electromagnetic current and zero-modes

Title: Lightfront projection of spin-1 electromagnetic current and zero-modes [22]

有从波函数计算到顶点的过程。

12.4 Covariant and light-front approaches to the ρ -meson electromagnetic form factors

Title: Covariant and light-front approaches to the ρ -meson electromagnetic form factors [19]

计算形状因子的过程。

12.5 Pion electromagnetic current in a light-front model

Title: Pion electromagnetic current in a light-front model [20]

计算pion的形状因子。

12.6 Unambiguous extraction of the electromagnetic form factors for spin-1 particles on the light-front

Title: Unambiguous extraction of the electromagnetic form factors for spin-1 particles on the light-front [21]

有具体计算过程的form factor。

12.7 Ward-Takahashi Identity on the Light Front

Title: Ward-Takahashi Identity on the Light Front [47]

计算光前下的Ward-Takahashi Identity

13 blfq method

13.1 Transverse momentum structure of strange and charmed baryons: a light-front Hamiltonian approach

Title: Transverse momentum structure of strange and charmed baryons: a light-front Hamiltonian approach [63]

在BLFQ框架下计算了 Λ 和 Λ_c 重子的领头扭度TMD（横动量依赖的部分子分布函数）。这些自旋1/2复合系统包含两个轻夸克（ u 和 d ）和一个 s/c 夸克。通过求解光前本征方程得到领头福克区的光前波函数，利用波函数的重叠积分计算TMD。同时还研究了不同极化下夸克在动量空间中的自旋密度分布，并在夸克模型框架下比较了奇异重子、粲重子和质子的TMD的差异，给出了TMD的概率论解释。

13.2 Transverse structure of the pion beyond leading twist with basis light-front quantization

Title: Transverse structure of the pion beyond leading twist with basis light-front quantization [62]

在BLFQ框架下研究了 π 介子的twist-3 TMD（横动量依赖的部分子分布函数）。twist-3 TMD并非独立的，可以通过运动方程（EOM）分解为twist-2项和genuine twist-3项。利用对角化光前QCD哈

密顿量得到的光前波函数（包含 $q\bar{q}$ 和 $q\bar{q}g$ 福克区间），并引入三维禁闭势，计算了这些TMD。同时还在twist-3层次计算了PDF，验证了求和规则的守恒。这是首次对包含两个福克区间干涉项的次领头扭度结构给出理论预言。

13.3 Heavy Quarkonium in a Holographic Basis

Title: Heavy Quarkonium in a Holographic Basis [39]

在BLFQ框架下研究重夸克偶素（粲偶素和底偶素）。文章实现了单胶子交换相互作用以及受光前全息（light-front holography）启发的禁闭势，以全息光前波函数（LFWF）作为基函数，通过对角化哈密顿量矩阵求解非微扰动力学，得到了粲偶素和底偶素的质量谱，并利用得到的LFWF计算了部分本征态的衰变常数和电荷形状因子，与实验测量和其他方法（格点QCD、DSE）的结果合理一致。

重点： $N_{\max}$ 截断的误差分析。BLFQ采用谐振子基矢，必须对横向截断 $N_{\max}$ （以及纵向截断 K ）进行系统的收敛性检验。文中给出了不同 $N_{\max}$ 下能谱的变化，用以评估基矢截断带来的系统性误差——这是BLFQ方法学中评估结果可靠性的关键步骤，也为后续工作（包括更高Fock区、更高 $N_{\max}$ 的外推）奠定了基础。

14 AI4Science

14.1 Heavy Quarkonium Spectrum and Decay Constants from a Neural-Network-Based Holographic Model

Title: Heavy Quarkonium Spectrum and Decay Constants from a Neural-Network-Based Holographic Model [61]

这是一篇用神经网络做 bottom-up AdS/QCD 模型的工作。作者没有使用传统的解析ansatz（如软墙模型中的二次dilaton），而是用多层感知机（MLP）学习dilaton场的导数 $\Phi'(z)$ 作为全息坐标的光滑函数，并施加 $\Phi(0) = 0$ 的边界条件以保证紫外一致性。通过对dilaton及其导数的自动微分生成全息势 $U(z)$ ，将对应的类薛定谔方程离散化对角化得到低能模式。本征值和bulk-to-boundary模式在近边界的行为给出质量谱和衰变常数。使用PDG上粲偶素和底偶素的实验数据进行训练，得到的非二次dilaton profile同时解决了重夸克偶素能谱和轻子衰变常数随径向激发单调压低的长期难题。拟合的RMS偏差分别为1.26%（粲偶素）和3.32%（底偶素）。文章展示了神经网络重建作为全息建模的灵活工具，并为未来扩展到其他道、格点约束或有限温度背景提供了基础。

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